

Nowcasting US Private Domestic Demand

Mixed-Frequency Dynamic Factor Model

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1 Data and model specification

We estimate a mixed-frequency Dynamic Factor Model for the United States. The target variable is quarterly *Real Final Sales to Private Domestic Purchasers* (TARGET), a measure of private domestic demand equal to personal consumption expenditures and gross private fixed investment. The sample covers the period from **January 1947 to March 2026**. Monthly indicators are obtained from FRED and BEA sources, while TARGET is observed at quarterly frequency. The objective of the project is to identify the main monthly drivers of U.S. private domestic demand. Since TARGET is largely composed of consumption and private investment,¹ the selected indicators were chosen to represent the main economic components underlying these aggregates.² The final dataset contains five monthly variables:

- **PCE**: captures household consumption, the largest component of private demand;
- **CAP**: nondefense capital goods shipments, proxying equipment investment;
- **RES**: residential construction spending, representing residential investment;
- **IP**: Industrial Production Index (FRED: INDPRO), a high-frequency coincident indicator of manufacturing and broader goods-producing activity;
- **PERMIT**: building permits, included as a leading indicator of future residential activity.

The model has a monthly latent factor that captures the common component of private domestic demand. CAP, RES, PCE and IP load contemporaneously on the factor f_t . Building permits are treated as a leading indicator and load on both f_{t+1} and f_{t+2} , reflecting the fact that construction permits typically anticipate future investment activity. The quarterly TARGET is linked to the monthly factor through the Mariano–Murasawa aggregation scheme discussed in class. All monthly variables are transformed as $100 \times \Delta \log(x_t)$. A preliminary Principal Component Analysis supports the single-factor assumption. The first principal component explains approximately 60% of the total variance of the standardized dataset, suggesting the presence of a dominant common business-cycle component.

2 Variable choice and model interpretation

The final specification was obtained through an extensive sequence of model-selection experiments. Several additional variables were tested, including consumer sentiment, construction employment, retail sales, manufacturing indicators and alternative investment measures. Variables were retained only when they improved the relationship between the estimated factor and TARGET while remaining economically meaningful. The final set of indicators was selected because it closely mirrors the composition of private domestic demand. PCE captures the consumption side of the economy, while CAP and RES represent the two main forms of private fixed investment covered by our specification: equipment and residential structures. IP (Industrial Production Index, FRED: INDPRO) provides a high-frequency supply-side signal of manufacturing and goods-producing activity, complementing the direct investment indicators and improving the factor's ability to track the cyclical component of private demand. Together, these variables provide direct information on the main sources of variation in TARGET. Building permits play a special role in the model. Unlike the other indicators, permits are not intended to measure current activity but rather future activity. For this reason, PERMIT is introduced through forward factor loadings on f_{t+1} and f_{t+2} , allowing the model to

¹Real Final Sales to Private Domestic Purchasers equals Personal Consumption Expenditures plus Gross Private Fixed Investment by national accounting identity. See Bureau of Economic Analysis, *National Income and Product Accounts*, Table 1.1.5 (Gross Domestic Product).

²The identification of monthly indicators with the components of Real Final Sales to Private Domestic Purchasers follows the expenditure-based decomposition of Gross Private Fixed Investment in Chapter 6 of the Bureau of Economic Analysis (BEA) National Income and Product Accounts (NIPA). Chapter 6 distinguishes equipment investment—linked to nondefense capital goods shipments (Table 6.B, MSM series)—from residential and nonresidential structures investment—linked to Census value-put-in-place surveys (Table 6.A, MCS series)—and identifies building permits as a leading indicator of residential activity (Table 6.A, line 39). The Industrial Production Index (FRED: INDPRO) was retained in the final specification as a high-frequency coincident indicator of goods-producing and manufacturing activity.

incorporate information about future residential investment before it appears in spending data. Several variables were ultimately excluded. NONRES (Nonresidential Construction Spending) was initially considered because it represents an important component of private investment. However, in our estimates it did not improve the ability of the factor to track TARGET and added little information beyond that already contained in the other investment indicators. For this reason, it was not retained in the final specification. Michigan Consumer Sentiment was also tested as a soft indicator but failed to improve model performance. The index combines backward-looking assessments, current conditions and long-horizon expectations, making its signal difficult to reconcile with a single factor representing contemporaneous private demand. The final specification achieves a correlation of approximately 0.84 between the estimated factor and TARGET, corresponding to an R^2 of approximately 0.71. The model-implied common component of TARGET also displays a strong correlation with the observed series. These results indicate that the estimated factor successfully captures the main cyclical movements of U.S. private domestic demand and provides an economically interpretable summary of consumption and investment dynamics.

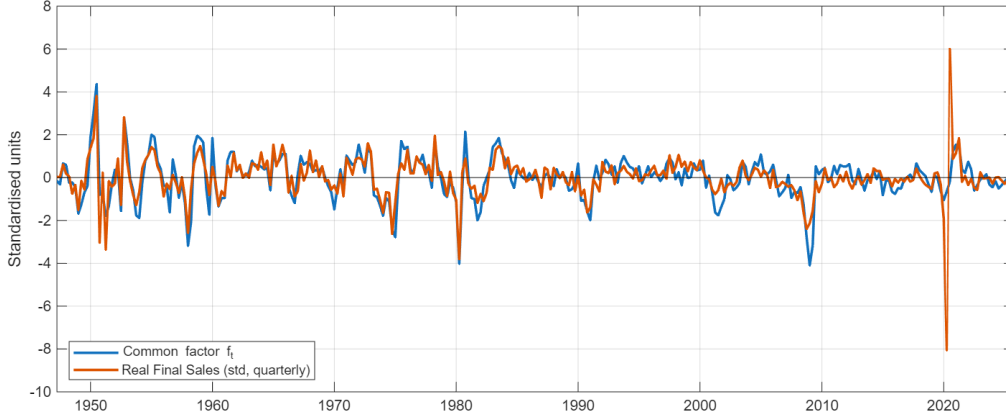


Figure 1: Common Factor vs Target

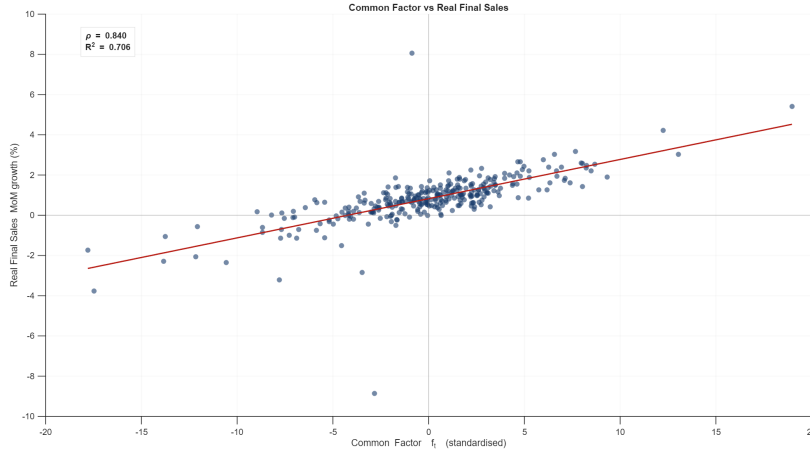


Figure 2: Common Factor vs Target, scatter plot

3 Forecast

Forecasts are obtained by iterating the state equation forward from the last available filtered state:

$$h_{T+s|T} = Fh_{T+s-1|T}.$$

The corresponding forecast of TARGET is obtained through the measurement equation. Since TARGET is observed at quarterly frequency while monthly indicators are available through March 2026, the model first produces a nowcast for the missing 2026Q1 observation. Forecasts are then generated from April 2026 onward by propagating the state vector forward without additional information.

Period	TARGET estimate
2026Q1	1.031 (nowcast)
2026Q2	0.756 (forecast)
2026Q3	0.625 (forecast)

Appendix: Experiment table

Table 1 summarizes the main model-selection experiments performed during the project. For each specification we evaluated parameter significance, factor dynamics, economic interpretability and the correlation between the estimated factor and TARGET. Following the selection criterion discussed in class, a modification was retained only if it improved the correlation between the common factor and the target variable relative to the current benchmark model.

Table 1: Summary of model-selection experiments

Run	Modification	Corr	Main finding	Decision
4	Baseline specification	0.651	Initial benchmark model with six monthly indicators.	Benchmark
6	+ PERMIT	0.685	Building permits provide useful leading information for residential investment.	Accepted
7	+ DGORDER	0.676	Durable goods orders introduce substantial idiosyncratic volatility.	Rejected
8	SENT (12-lag structure)	0.369	Sentiment generates excessively persistent factor dynamics and weakens the hard-data signal.	Rejected
9	−PCE	0.667	PCE remains essential for identifying the consumption component of private demand.	Rejected
10	+ RSBPBSS	0.669	Construction retail spending weakens the common investment signal.	Rejected
11	+ CONEMP	0.615	Employment reacts too slowly to business-cycle turning points.	Rejected
12	RES distributed lag	0.673	Improves likelihood but reduces contemporaneous correlation with TARGET.	Rejected
13	+ RSXFS	0.440	Strong collinearity with PCE causes identification problems.	Rejected
14	+ IPMAN	0.660	Manufacturing activity is not perfectly aligned with private demand dynamics.	Rejected
17	−IPP	0.719	Removing the smooth IPP (Intellectual Property Products) series strengthens cyclical factor dynamics.	Accepted
18	−NONRES	0.733	NONRES contributes more noise than signal and lowers factor quality.	Best model (Phase I)
19	+ SENT (contemporaneous)	0.720	Consumer sentiment is largely absorbed by its idiosyncratic component.	Rejected
24	Final specification: CAP, RES, PCE, IP, PERMIT(f_{t+1}, f_{t+2})	0.840	Final model. Building permits are included as a leading indicator through two forward factor loadings, while IP (Industrial Production Index, FRED: INDPRO) provides a high-frequency coincident signal of manufacturing and goods-producing activity. The specification achieves the highest correlation with TARGET and provides the most complete economic representation of private domestic demand.	Final model